SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Customer Relationship Management System

Authors: Shweta, Avaneesh N, Vishwas V Angadi, Krishna Pai
Version 2.2 | 04-Nov-2026 | Status: Revised Draft

Document Control

Field	Details
Project	Customer Relationship Management (CRM)
Document	Software Requirements Specification
Authors	Shweta, Avaneesh N, Vishwas V Angadi, Krishna Pai
Scope baseline	GitHub Project Management issues visible in the supplied screenshots (#6–#34)
Status	Revised Draft — team review and instructor approval pending

Approvals

Role	Name
Team Member	Shweta
Team Member	Avaneesh N
Team Member	Vishwas V Angadi
Team Member	Krishna Pai
Instructor / Reviewer	To be completed

Table of Contents

- 1. Introduction
- 2. Overall Description
- 3. External Interface Requirements
- 4. System Features (Detailed)
- 5. Non-Functional Requirements (Detailed)
- 6. Quality Attributes and Acceptance Tests
- 7. System Models and Diagrams
- 8. Requirements Traceability Matrix

1. Introduction

1.1 Purpose

This SRS defines the functional and non-functional requirements for the CRM application developed by Team 12. It is based on the project scope represented by the team's GitHub issues and the features implemented in the application. It is the shared baseline for architecture/design, testing and project evaluation.

1.2 Scope

The CRM supports staff login/logout, authentication and password validation, role-based access control, user account management, lead creation and management, lead-to-customer conversion, customer profile management and assignment, interaction tracking, follow-up management, feedback, retention, sales, campaign management, dashboards/reports and audit logging. Project quality work includes GitHub Actions and pull-request checks, system/regression testing and documentation. Non-functional concerns include performance, persistence, modularity, safe error handling, secret protection and protected pages/APIs.

1.3 Audience

Developers, testers, project team members, instructor/evaluators and future maintainers.

1.4 Definitions

Term	Meaning
CRM	Customer Relationship Management
RBAC	Role-Based Access Control
API	Application Programming Interface
FR / NFR	Functional Requirement / Non-Functional Requirement
SRS / SAD / STP	Software Requirements Specification / Software Architecture and Design / Software Test Plan
RTM	Requirements Traceability Matrix
CI	Continuous Integration

2. Overall Description

2.1 Product Perspective

The CRM is a web application. Users interact through a browser-based frontend. The frontend communicates with backend APIs. The backend applies authentication, role authorization and business rules before accessing persistent CRM data. GitHub Actions supports automated repository checks.

2.2 Major Product Functions

- Authentication and user management: login/logout, credential validation, RBAC and account management.
- Lead and customer management: create/manage leads, convert leads, maintain customer profiles and assign customers to Sales Staff.
- Customer engagement: track interactions, manage follow-ups, feedback and retention activities.
- Business operations: manage sales and campaigns.
- Oversight: dashboards, reports and audit logging.
- Engineering workflow: GitHub Actions/PR checks, system/regression testing and project documentation.

2.3 User Roles and Characteristics

Role	Description
Admin	Performs administrative operations and oversight available to the Admin role.
Sales Staff	Works with leads, customers assigned to the staff member and permitted customer/business activities.
Support Staff	Performs customer engagement and support activities permitted by the implemented role permissions.

The exact permissions for each role are those enforced by the implemented application. The UI must not be treated as the sole security boundary; protected API operations must enforce access.

2.4 Operating Environment

The application runs in the team's configured development environment and is accessed through a modern desktop browser. The project uses its configured frontend, backend and database services.

2.5 Constraints

- Requirements are limited to the CRM project scope shown in the supplied GitHub Project screenshots.
- Authentication and role authorization must be enforced for protected operations.
- CRM records must persist in the configured database.
- The application should retain modular frontend, backend and data responsibilities.
- No unlisted external integrations are assumed.

2.6 Assumptions and Dependencies

- The source repository and GitHub Project board are available.
- The configured backend and database are available during use and testing.
- Test users for the implemented roles can be provisioned.
- GitHub Actions and pull-request checks are enabled in the repository.

3. External Interface Requirements

3.1 User Interfaces

The user interface provides authentication screens and role-appropriate navigation. It provides forms, lists and views for the CRM modules listed in Section 4. User actions should receive understandable success, validation and error feedback.

3.2 Hardware Interfaces

No specialized hardware interface is required by the project scope.

3.3 Software Interfaces

Interface	Requirement
Frontend / Backend	The frontend communicates with backend APIs to authenticate users and perform CRM operations.
Backend / Database	The backend reads and writes CRM records through the configured database/model layer.
GitHub	The repository uses GitHub Actions and pull-request checks as represented by project issue #26.

3.4 Communications

The frontend and backend exchange application requests and responses using the configured API protocol. Deployed environments should use HTTPS. Error responses must not disclose credentials, tokens or internal implementation details.

4. System Features (Detailed)

4.1 Authentication and User Management

Requirement ID	Requirement (shall...)	Acceptance Criteria	Project Issue
CRM-F-001	The system shall allow registered staff users to log in and log out.	Valid credentials permit access; logout ends access to protected functions.	#6
CRM-F-002	The system shall validate authentication credentials and password input.	Valid credentials are accepted and invalid credentials are rejected safely.	#7
CRM-F-003	The system shall enforce role-based access control for Admin, Sales Staff and Support Staff.	Each role can perform only operations permitted by the implemented role policy.	#8
CRM-F-004	The system shall support authorized user account management.	Authorized account changes are saved and reflected in the	#9

		user records.	

4.2 Lead and Customer Management

Requirement ID	Requirement (shall...)	Acceptance Criteria	Project Issue
CRM-F-005	The system shall support lead creation and management.	Authorized users can create, view and update lead records.	#11
CRM-F-006	The system shall support conversion of an eligible lead into a customer.	Conversion creates/links a customer and updates the lead appropriately.	#12
CRM-F-007	The system shall support customer profile management.	Authorized profile changes are saved and displayed.	#13
CRM-F-008	The system shall support customer assignment to Sales Staff.	The assignment is saved and reflected in staff/customer views.	#14

4.3 Customer Engagement and Support

Requirement ID	Requirement (shall...)	Acceptance Criteria	Project Issue
CRM-F-009	The system shall track customer interactions.	An interaction is saved with its associated customer and staff context.	#16
CRM-F-010	The system shall support customer follow-up management.	Authorized users can create and manage follow-up records.	#17
CRM-F-011	The system shall support customer feedback management.	Feedback records can be created, viewed and updated by permitted users.	#18
CRM-F-012	The system shall support customer retention management.	Retention activities can be recorded and managed.	#19

4.4 Sales, Campaigns, Dashboards and Audit

Requirement ID	Requirement (shall...)	Acceptance Criteria	Project Issue
CRM-F-013	The system shall support sales management.	Authorized sales records can be created and maintained.	#22
CRM-F-014	The system shall support campaign management.	Campaign records can be managed through the implemented campaign feature.	#23
CRM-F-015	The system shall provide dashboards and reports.	Dashboard and report views display CRM summary information.	#24
CRM-F-016	The system shall provide audit logging for relevant application activities.	Relevant events are recorded and available to authorized reviewers.	#25
CRM-F-017	The repository shall run configured GitHub Actions and pull-request checks.	The configured workflow runs for applicable repository/PR events.	#26
CRM-F-018	The team shall perform system and regression testing.	Planned system/regression tests are executed and results recorded.	#27
CRM-F-019	The team shall complete project documentation.	Project documentation reflects the implemented application.	#28

5. Non-Functional Requirements (Detailed)

Req ID	Requirement	Acceptance Criteria / Measurement	Category	Issue
CRM-NF-001	The dashboard and list views shall load within the agreed project performance target.	Measure representative loading times and record test conditions; target is agreed by the team.	Performance	#29
CRM-NF-002	Successfully saved CRM records shall remain available after page refresh and application restart.	Create/update records, refresh/restart and verify persistence.	Reliability / Persistence	#30
CRM-NF-003	The application shall maintain a modular and maintainable structure.	Review separation of frontend, backend and data/model responsibilities.	Maintainability	#31
CRM-NF-004	The system shall handle errors safely and protect application secrets.	Invalid operations return safe errors; secrets are not exposed in responses or logs.	Security / Reliability	#32
CRM-NF-005	The system shall protect restricted pages and APIs from unauthorized access.	Unauthenticated or unauthorized access is rejected by the application/API.	Security	#34

5.1 Security

5.1.1 Security Objectives

- SO-1 — Access control: ensure that protected CRM pages and API operations are available only to authenticated users with the required role.
- SO-2 — Data and secret protection: prevent unauthorized changes to CRM records and avoid exposing credentials, tokens or application secrets.

5.1.2 Security Requirements

Security Req ID	Requirement (shall...)	Priority	Acceptance / Verification
CRM-SR-001	The system shall authenticate users before granting access to protected functions.	High	Authentication test; protected page/API access test.
CRM-SR-002	The backend shall enforce role-based authorization for protected operations.	High	Attempt permitted and forbidden operations for each role.
CRM-SR-003	The system shall validate credentials and protect password/secret values from disclosure.	High	Invalid/valid login tests; inspect responses and logs for secrets.
CRM-SR-004	The system shall return safe error responses that do not expose internal details or secrets.	High	Submit invalid requests and inspect error output.
CRM-SR-005	The system shall prevent unauthorized modification of protected CRM records.	High	Attempt unauthorized create/update/delete operations and verify rejection.

These security requirements describe security behavior developed in the CRM. They are not represented as separate GitHub issues; they are verified through the related authentication, RBAC, safe-error and page/API protection work.

6. Quality Attributes and Acceptance Tests

Quality Attribute	Acceptance Test
Functional completeness	Verify all functional requirements in Section 4 using the corresponding test cases.
Performance	Measure dashboard and list loading under documented test conditions (#29).
Persistence	Verify saved CRM records after refresh and application restart (#30).
Maintainability	Review modular organization and separation of concerns (#31).
Safe error handling	Verify safe errors and no secret disclosure (#32).
Security	Verify authentication, role permissions and protected pages/APIs (#7, #8, #34).

Acceptance status must be based on executed tests and recorded evidence. This SRS does not presume that tests have passed.

7. System Models and Diagrams

The following use-case diagrams show the principal interactions for the defined CRM roles. They represent the project scope; actual access to each function is determined by the implemented role permissions.

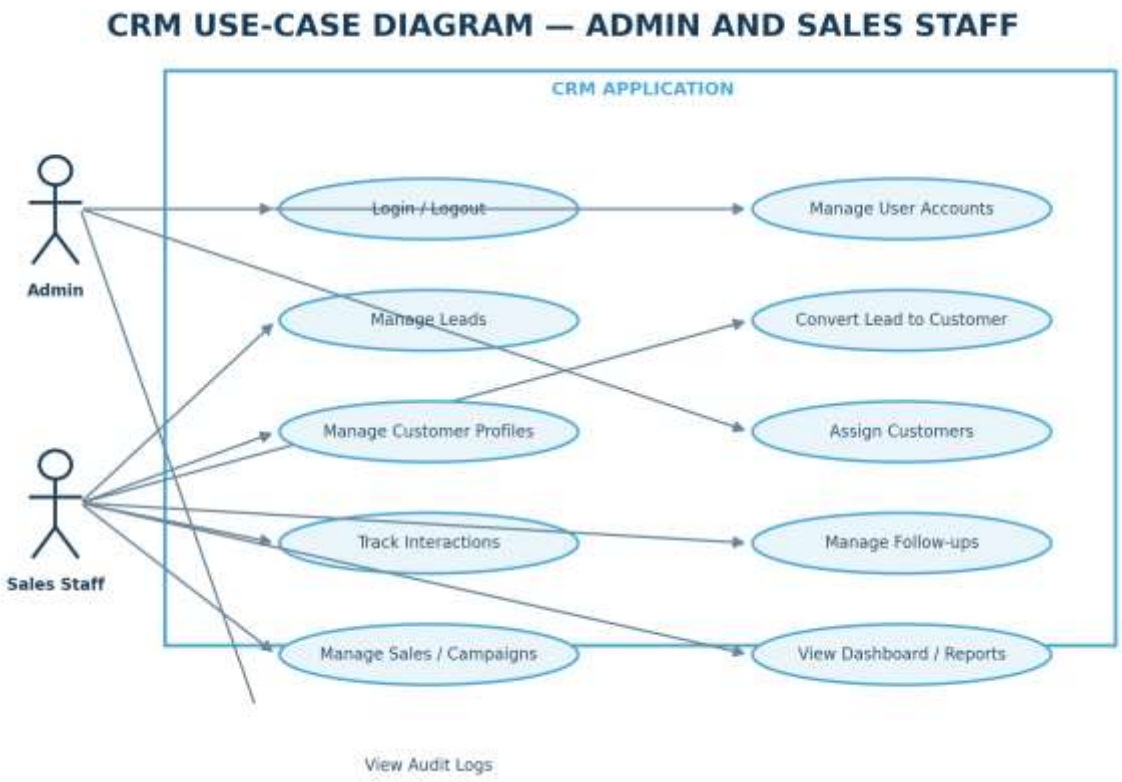


Figure 1. UML use-case diagram — Admin and Sales Staff.

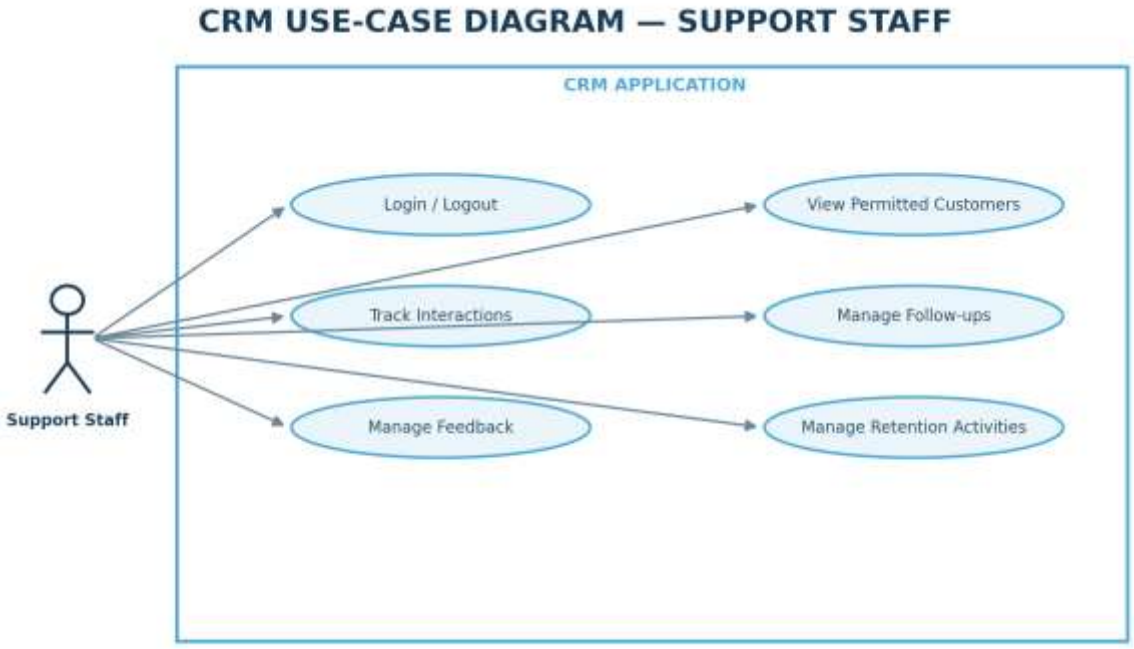


Figure 2. UML use-case diagram — Support Staff.

8. Requirements Traceability Matrix (RTM)

Req ID	Requirement Summary	Source
CRM-F-001	The system shall allow registered staff users to log in and log out.	#6
CRM-F-002	The system shall validate authentication credentials and password input.	#7
CRM-F-003	The system shall enforce role-based access control for Admin, Sales Staff a…	#8
CRM-F-004	The system shall support authorized user account management.	#9
CRM-F-005	The system shall support lead creation and management.	#11
CRM-F-006	The system shall support conversion of an eligible lead into a customer.	#12
CRM-F-007	The system shall support customer profile management.	#13
CRM-F-008	The system shall support customer assignment to Sales Staff.	#14
CRM-F-009	The system shall track customer interactions.	#16
CRM-F-010	The system shall support customer follow-up management.	#17
CRM-F-011	The system shall support customer feedback management.	#18
CRM-F-012	The system shall support customer retention management.	#19
CRM-F-013	The system shall support sales management.	#22
CRM-F-014	The system shall support campaign management.	#23
CRM-F-015	The system shall provide dashboards and reports.	#24
CRM-F-016	The system shall provide audit logging for relevant application activities.	#25
CRM-F-017	The repository shall run configured GitHub Actions and pull-request checks.	#26
CRM-F-018	The team shall perform system and regression testing.	#27
CRM-F-019	The team shall complete project documentation.	#28
CRM-NF-001	The dashboard and list views shall load within the agreed project performan...	#29
CRM-NF-002	Successfully saved CRM records shall remain available after page refresh an...	#30
CRM-NF-003	The application shall maintain a modular and maintainable structure....	#31
CRM-NF-004	The system shall handle errors safely and protect application secrets....	#32
CRM-NF-005	The system shall protect restricted pages and APIs from unauthorized access...	#34
CRM-SR-001	The system shall authenticate users before granting access to protected fun...	Related issues #7, #8, #32, #34
CRM-SR-002	The backend shall enforce role-based authorization for protected operations...	Related issues #7, #8, #32, #34
CRM-SR-003	The system shall validate credentials and protect password/secret values fr...	Related issues #7, #8, #32, #34
CRM-SR-004	The system shall return safe error responses that do not expose internal de...	Related issues #7, #8, #32, #34
CRM-SR-005	The system shall prevent unauthorized modification of protected CRM records...	Related issues #7, #8, #32, #34